

Project Proposal: Encoding a Target Axial Response in Binary-Amplitude Fresnel Zone Plates via Radial Binary Optimization

1 Background and Motivation

A standard Fresnel zone plate (FZP) is a set of concentric transparent/opaque half-wave zones: at the focus the open zones add in phase, and the flat plate behaves as a focusing element. The zone boundaries are fixed by a single design rule, $r_n = \sqrt{n\lambda f + n^2\lambda^2/4}$ (paraxial $r_n \approx \sqrt{n\lambda f}$), so a conventional FZP is committed to one focus from the outset.

This project asks a narrower question: since the binary open/opaque pattern is what produces the field, can we choose that pattern to realize a *specified* axial intensity rather than a single textbook focus? We treat the FZP as a binary optical-field encoder. The radius is discretized into M rings with binary states $a_m \in \{0, 1\}$, and we optimize the vector $\mathbf{a}$ so that the on-axis intensity $\hat{I}_a(z)$ approximates a target profile $I_{\text{target}}(z)$. This reframes the FZP from a fixed diffractive lens into a small, transparent design problem while staying entirely within binary-amplitude fabrication. The scope is deliberately modest: a clean forward model, a light-weight discrete optimizer, and the same optical bench and standard-FZP control already established in our baseline (Scheme 1).

2 Proposed Approach

Design. We keep the verified baseline parameters: wavelength $\lambda = 532$ nm, outer radius $R = 2$ mm, nominal focus $f_0 = 150$ mm, giving outermost zone width $\Delta r_N \approx 20 \mu\text{m}$ and $N \approx 50$ Fresnel zones. The radius $[0, R]$ is divided into M rings, each at least the minimum manufacturable width $\Delta r_{\min} \sim 20 \mu\text{m}$, set by the same constraint $\Delta r_N \approx \lambda f / (2R)$. Each ring carries a binary aperture variable a_m (1 = open, 0 = opaque), and the transmittance is $T_B(\rho) = \sum_m a_m \Pi_m(\rho)$. We study three targets: (i) single focus, a Gaussian centered at $f_0 = 150$ mm (a sanity check against the standard FZP); (ii) *dual focus*, two Gaussians at $f_1 = 135$ mm and $f_2 = 165$ mm; (iii) *extended depth of focus*, a super-Gaussian (order $p = 4$) of half-range $\Delta f = 15$ mm about 150 mm.

Simulation. Each ring is modeled as a phasor source. We precompute its complex contribution at axial plane z_j ,

$$C_{mj} = \frac{2\pi e^{ikz_j}}{i\lambda z_j} \int_{r_{m-1}}^{r_m} \exp\left(\frac{ik\rho^2}{2z_j}\right) \rho d\rho,$$

so the field is $U_a(z_j) = \sum_m a_m C_{mj}$ and $I_a(z_j) = |U_a(z_j)|^2$; optimization simply chooses which phasors add. Because C_{mj} is precomputed, evaluating a pattern is a fast matrix product and a single ring flip updates U_a incrementally. We minimize

$$J(\mathbf{a}) = \sum_j w_j [\hat{I}_a(z_j) - I_{\text{target}}(z_j)]^2 + \alpha S_{\text{sidelobe}} + \beta C_{\text{edge}} + \gamma C_{\min},$$

where S_{sidelobe} penalizes out-of-band axial intensity, $C_{\text{edge}} = \sum_m |a_m - a_{m-1}|$ penalizes pattern fragmentation, and $C_{\min}$ penalizes ring runs shorter than $\Delta r_{\min}$. No heavy machine learning is needed: we use binary local search / simulated annealing (flip one ring; accept if $\Delta J < 0$, else with probability $e^{-\Delta J/T}$; cool T to convergence), initialized from a standard FZP or at random. Full Fresnel/angular-spectrum propagation, $U(x, y, z) = P_z\{U_0 T(x, y)\}$, then validates the final patterns in 2-D before fabrication.

Fabrication. The optimized binary patterns and a standard FZP control are written as chrome-on-glass (or high-resolution film) masks at the same $20 \mu\text{m}$ linewidth, so every design is manufacturable and directly comparable.

Testing. We reuse the Scheme-1 bench unchanged: a 532 nm laser, beam expander/collimator, the zone plate, and a CMOS camera translated along the axis over $z = 120$ – 180 mm (coarse 1–2 mm, fine 0.2–0.5 mm). For each design we record the axial peak $I_{\max}(z) = \max_{xy} I$ and spot FWHM(z), and report: the target-match error $E_{\text{exp}} = \sum_j [\hat{I}_{\max}(z_j) - I_{\text{target}}(z_j)]^2$; the effective depth of focus $\text{DOF}_\eta = \text{length}\{z : I_{\max}(z) \geq \eta I_{\max, \text{peak}}\}$ ($\eta = 0.5, 0.7$); and defocus tolerance in imaging via contrast $C = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$ and mean edge gradient $G = \langle |\nabla I| \rangle$. Repeating each scan K times gives $u_I(z_j)$, and the focal uncertainty follows $u(f_{\text{exp}}) = \sqrt{u_z^2 + u_{\text{fit}}^2 + u_{\text{align}}^2}$ with $\delta f / f \approx 2 \delta r / r - \delta \lambda / \lambda$ (a 1% pattern scaling gives $\sim 2\%$ focal error). The standard FZP serves throughout as the control.

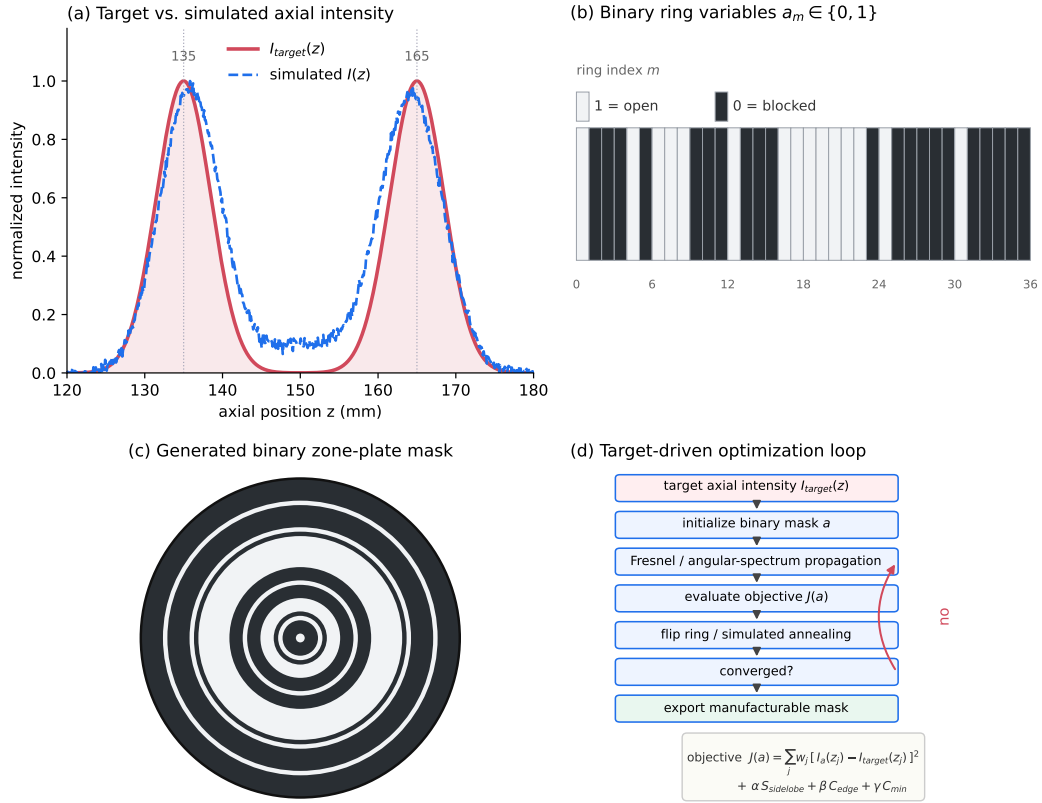


Figure 1: Radial binary optimization as a target-encoding mechanism. **(a)** A target axial intensity (here a dual focus at 135 and 165 mm) and the response of an optimized pattern. **(b)** The aperture is encoded as a vector of binary ring states $a_m \in \{0, 1\}$, which **(c)** generates a concentric binary mask. **(d)** The ring states are optimized against the target by binary local search / simulated annealing, minimizing the objective $J(a)$.

3 Expected Outcomes

We expect to show, on the same bench, that a binary ring vector can be encoded for a chosen axial response: a single focus matching the control, a measurable dual-focus split at 135 and 165 mm, and an extended-DOF design with a flatter $I_{\text{max}}(z)$ plateau. The central, honest result is an energy-conservation tradeoff: a finite aperture has finite energy, so distributing light to two foci or across a focal range *lowers* the single-point peak. We therefore aim to quantify the relationship high peak $\leftrightarrow$ short DOF versus extended DOF / multi-focus $\leftrightarrow$ lower peak and higher sidelobes, using E_{exp} , DOF_η , and sidelobe levels as evidence, and to report it with full uncertainty budgets rather than as a single “best” design.

Honest caveats. Optimized patterns are irregular and less interpretable than a textbook FZP, more sensitive to fabrication error, and the discrete search can settle in local optima; the regularizers (C_{edge} , C_{min}) and annealing mitigate but do not remove these. As with any FZP, zero-order light, multiple foci, and chromatic aberration remain present.

4 Tentative Timeline

Both schemes share one optical bench, and the standard FZP is the common control.

Week	Activity
1	Build forward model; precompute ring contributions C_{mj} ; validate vs. standard FZP
2	Implement objective $J(\mathbf{a})$ and annealing/local search; tune regularizers
3	Optimize single-, dual-, and extended-DOF targets; 2-D propagation check
4	Finalize patterns; prepare and order masks (optimized designs + control)
5	Receive masks; align bench; calibrate axial scan and camera
6	Axial-scan measurements: $I_{\text{max}}(z)$, FWHM, E_{exp} , DOF_η
7	Imaging / defocus-tolerance tests (C , G); repeat runs for $u_I(z_j)$
8	Uncertainty budget; compare to control; write-up and figures